Abstract Linear Algebra

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Please pick up a copy of the handouts.

Try those warm-up problems.

Next up:

- Read Ch 1 sections 5-7 of *LADW,* and Ch 2 sections 1-3
- Come to office hours! Today 3:30-4:30pm in Eckhart 207A; and small group office hours.
- Problem session 3-5 on Friday in Ryeerson 177
- HW1 due end of week
- Quiz 1 & 2 on Friday

Today in class:

- Explain homework procedures
- Discuss warm-ups
- Overview content of chapters 3 and beyond
- Discuss formal definitions of groups, fields, and vector spaces

Warm-up 1

a) Compute in $\mathbb{Z}/5\mathbb{Z}$:

$4 + 3 =$

$2 - 3 =$

$3 \times 4 =$

b) Compute all of the scalar multiples of $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ in $(\mathbb{Z}/5\mathbb{Z})^2$

Warm-up 1

a) Compute in $\mathbb{Z}/5\mathbb{Z}$:

$4 + 3 =$ 2

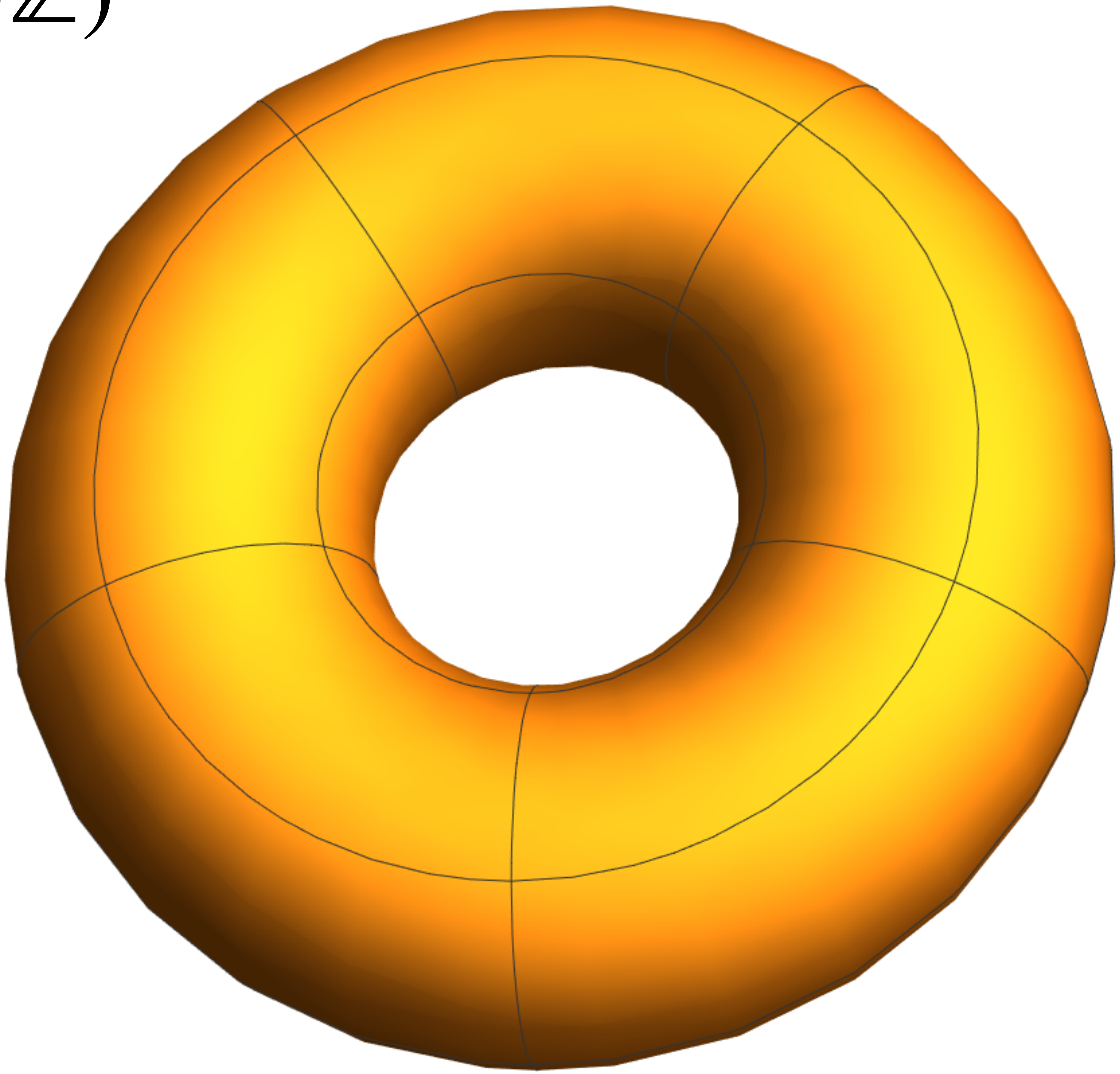
$2 - 3 =$ 4

$3 \times 4 = = 3 \times (-1) = -3 = 2$

b) Compute all of the scalar multiples of $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ in $(\mathbb{Z}/5\mathbb{Z})^2$

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad \begin{pmatrix} 2 \\ 4 \end{pmatrix} \quad \begin{pmatrix} 3 \\ 1 \end{pmatrix} \quad \begin{pmatrix} 4 \\ 3 \end{pmatrix}$$

$(\mathbb{Z}/5\mathbb{Z})^2$



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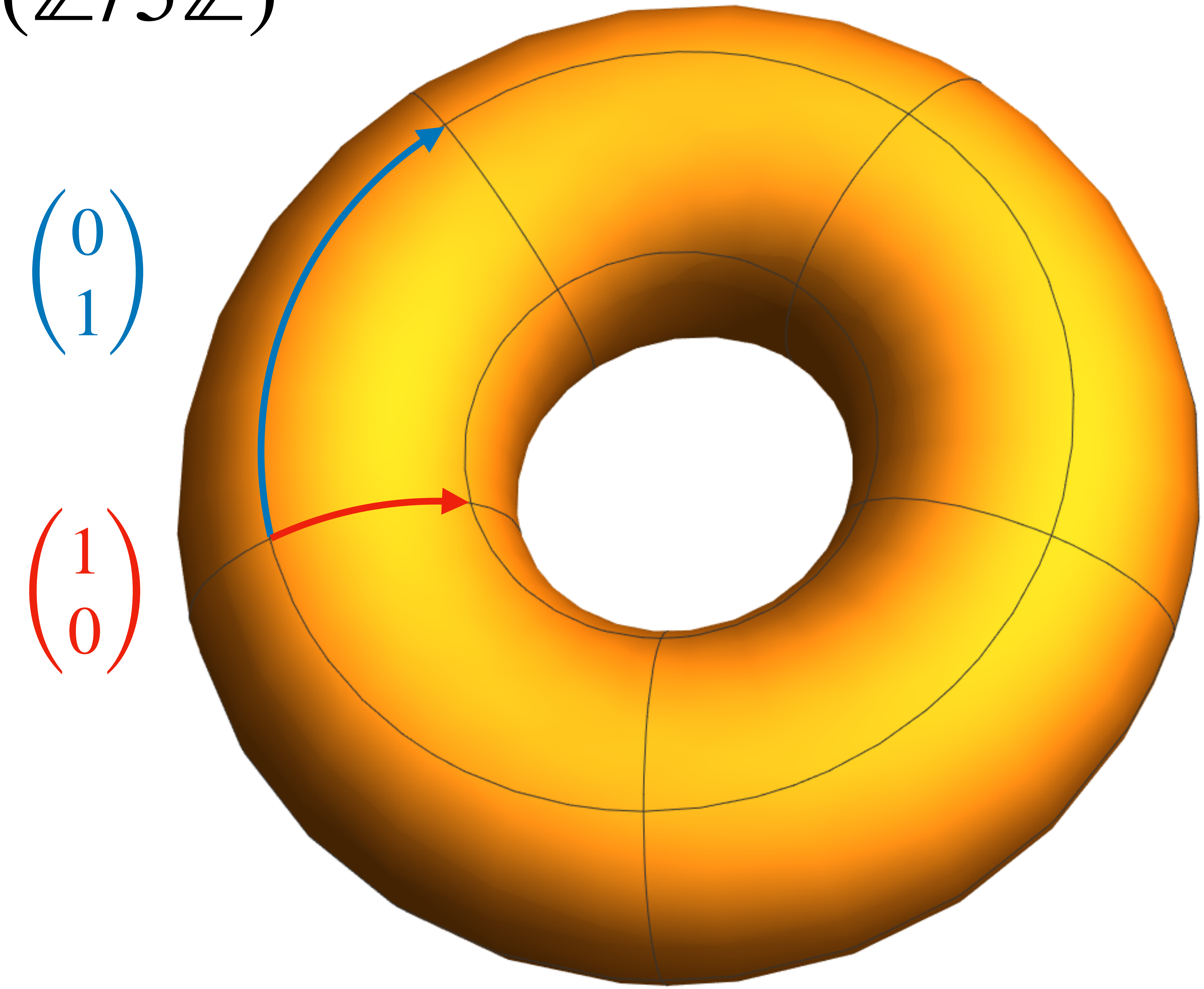

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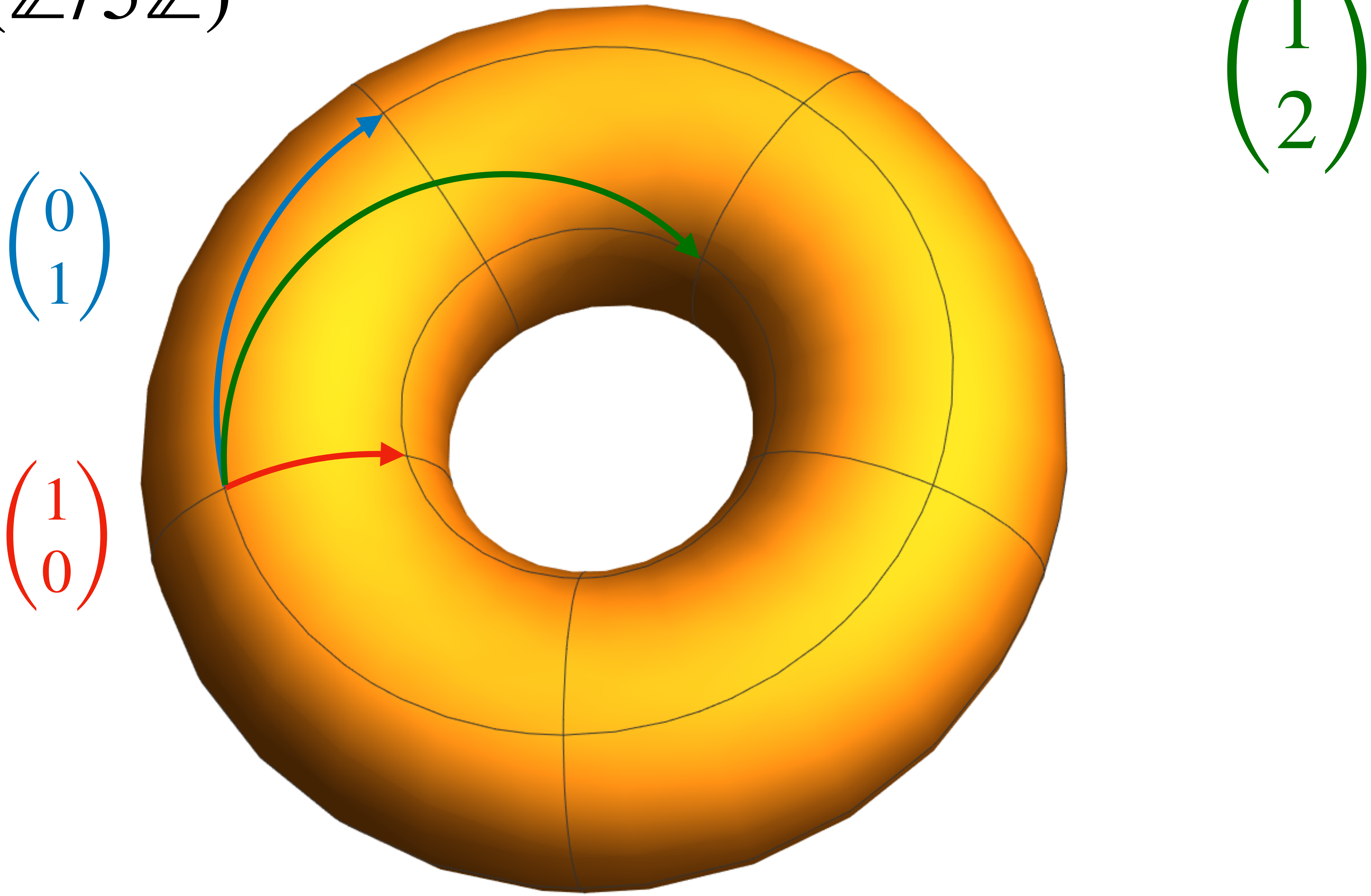

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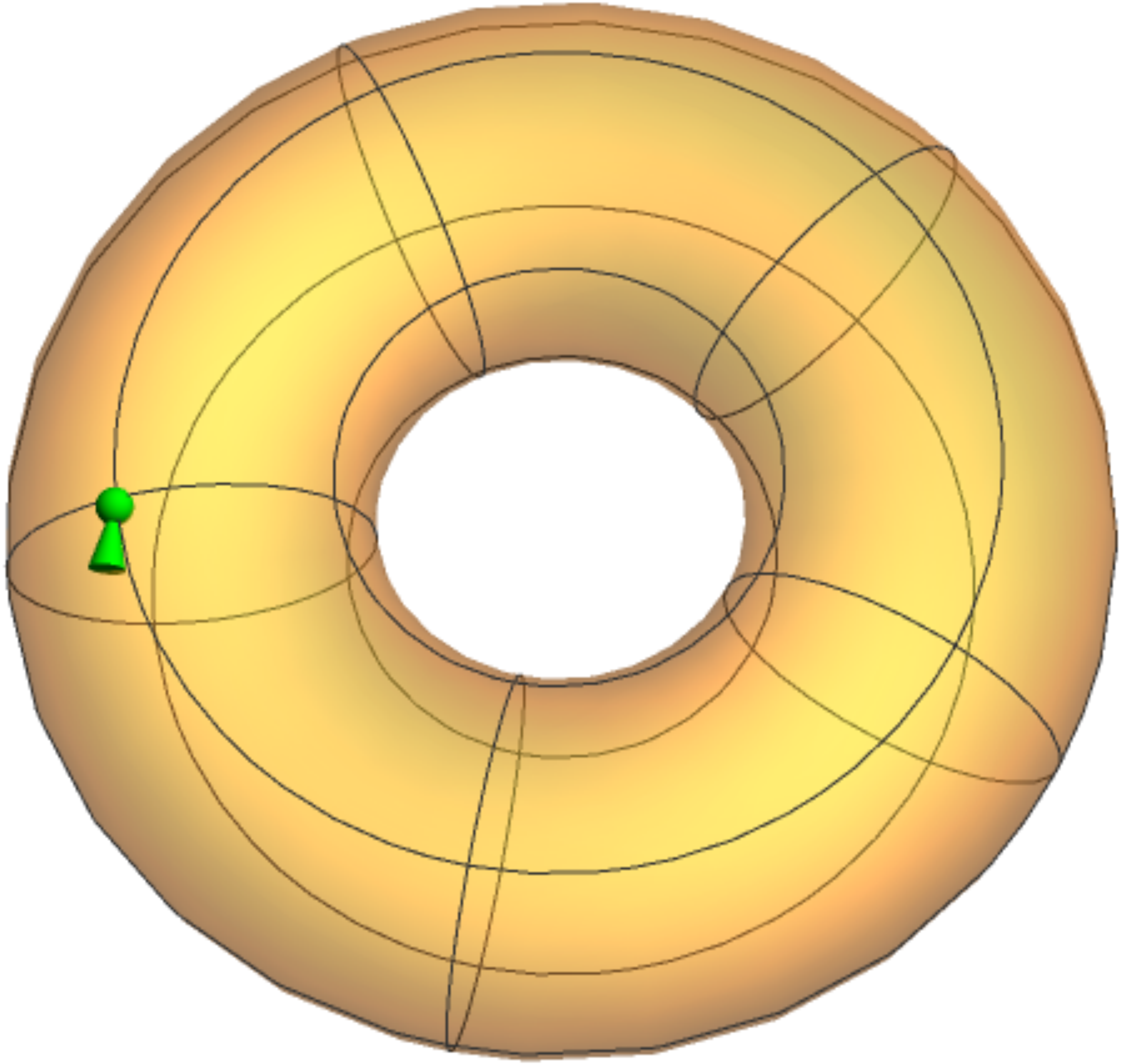

$$\begin{pmatrix}0\\0\end{pmatrix} \quad \begin{pmatrix}1\\2\end{pmatrix} \quad \begin{pmatrix}2\\4\end{pmatrix} \quad \begin{pmatrix}3\\1\end{pmatrix} \quad \begin{pmatrix}4\\3\end{pmatrix}$$

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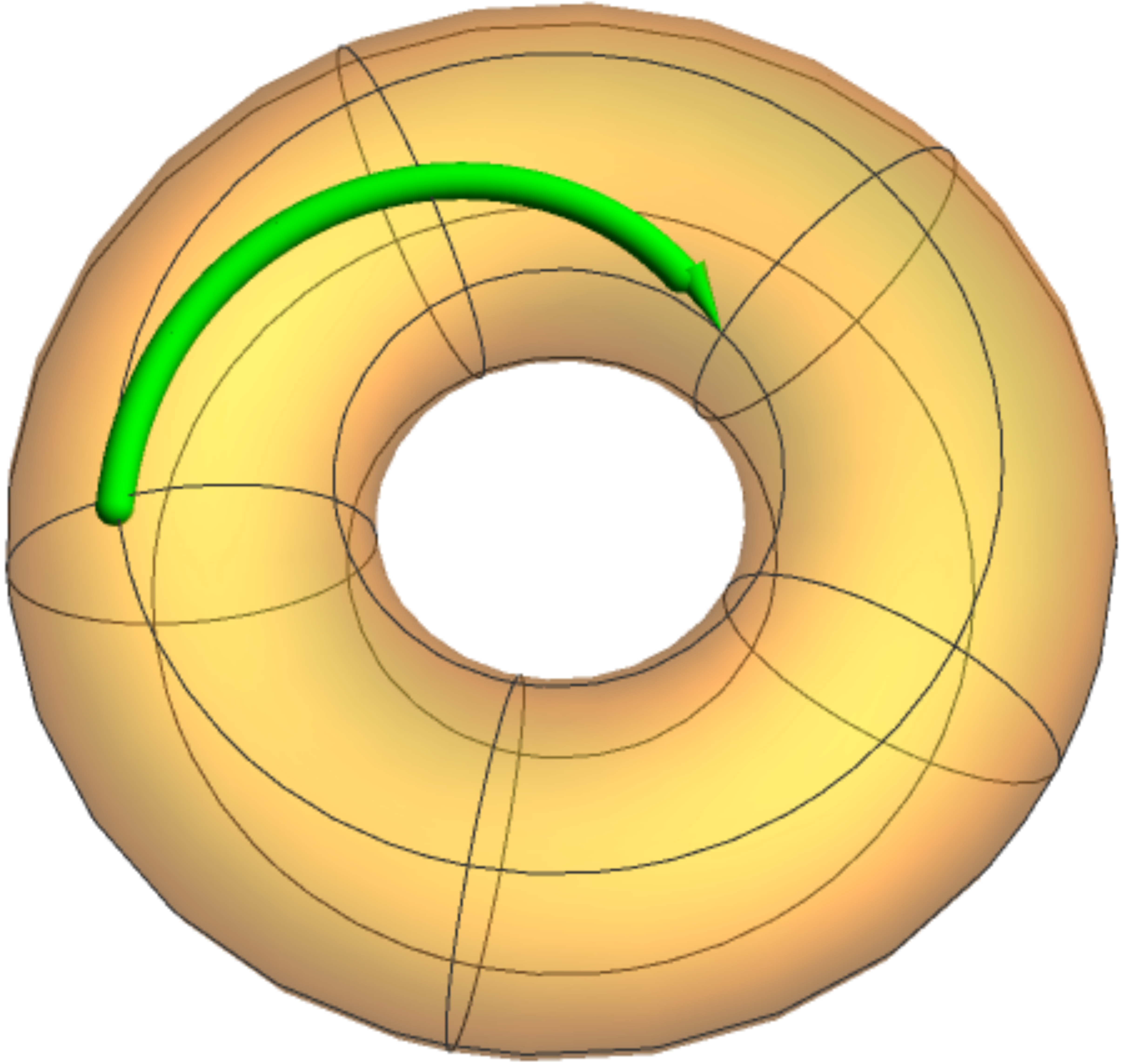

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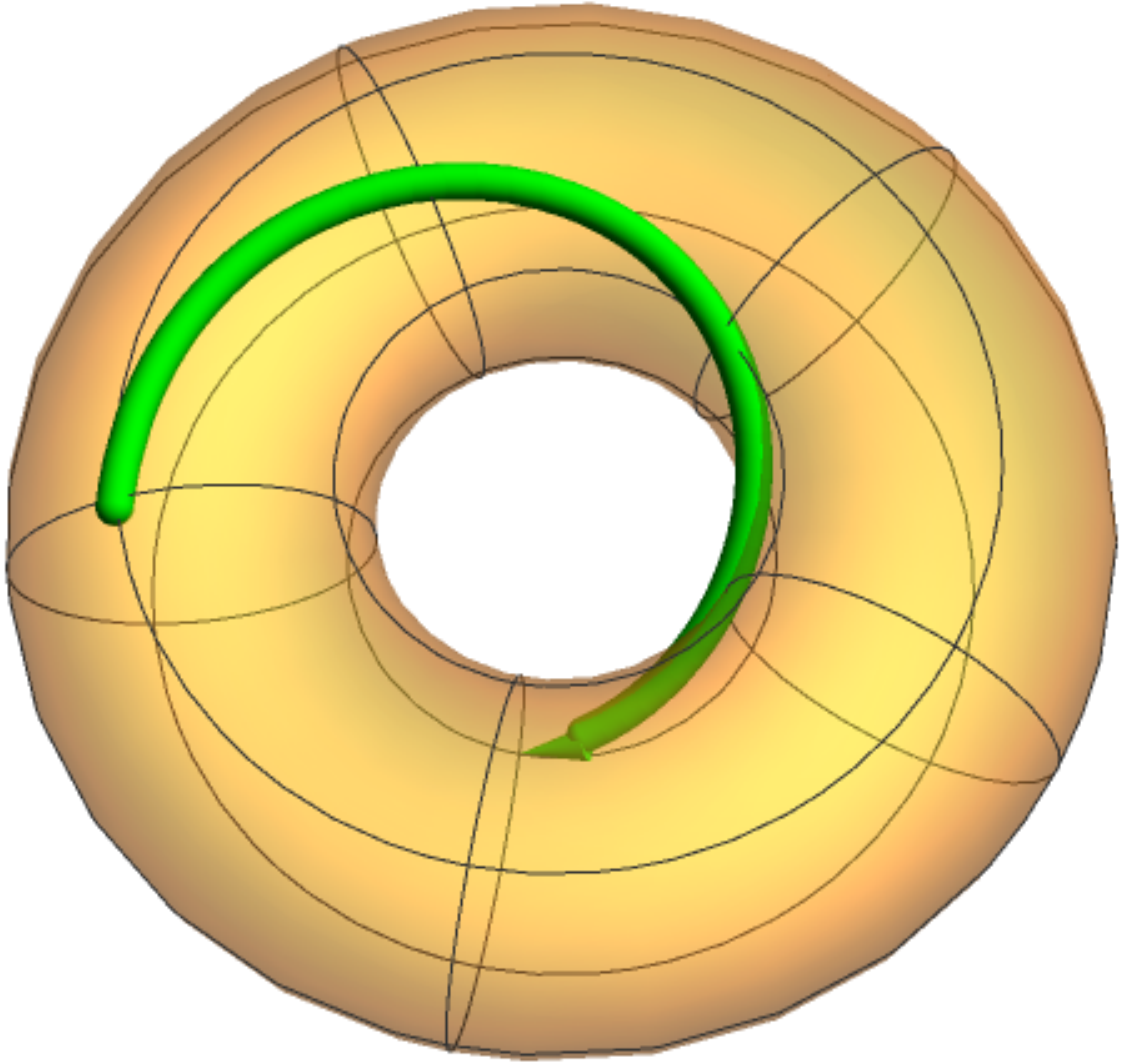

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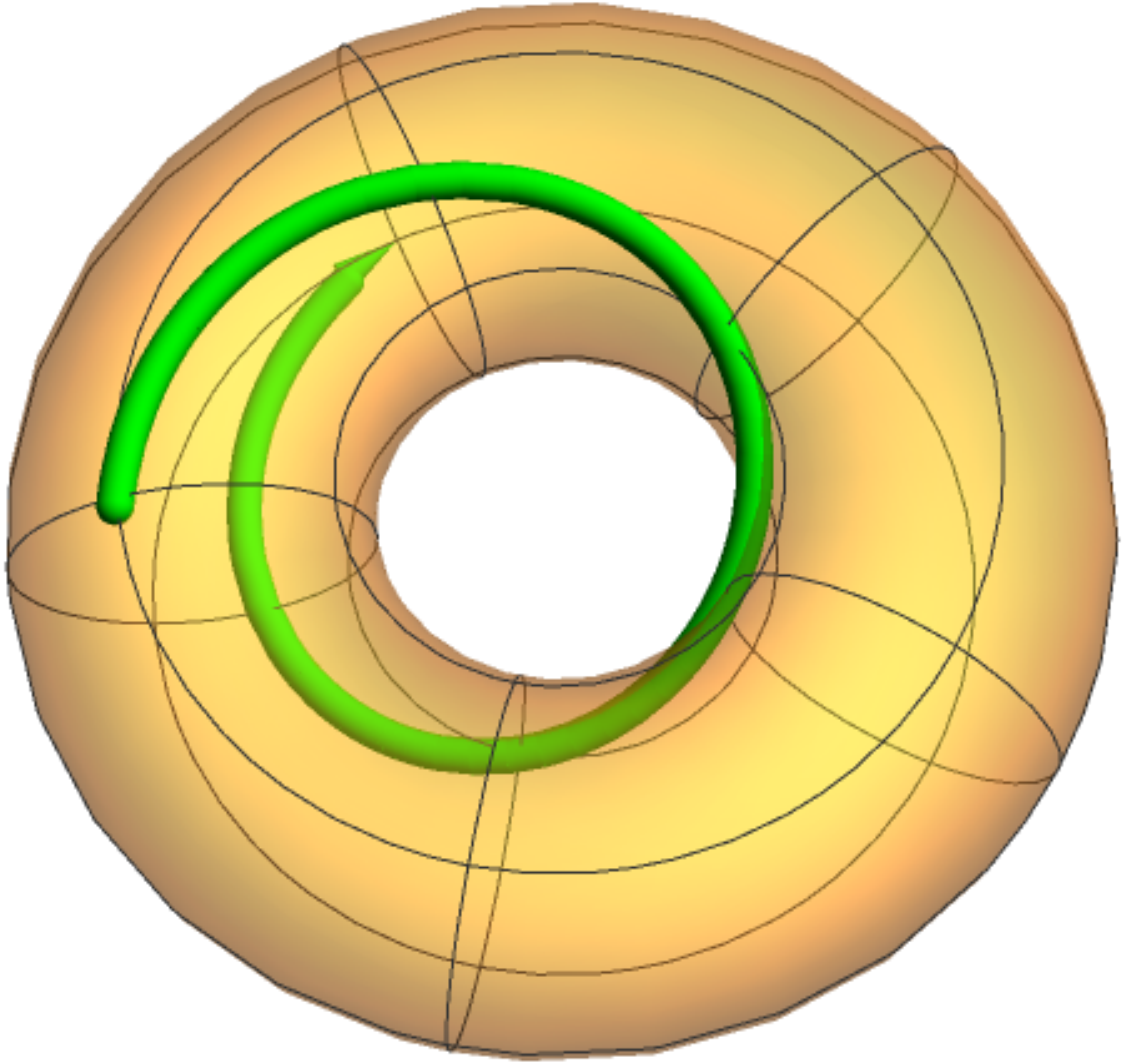

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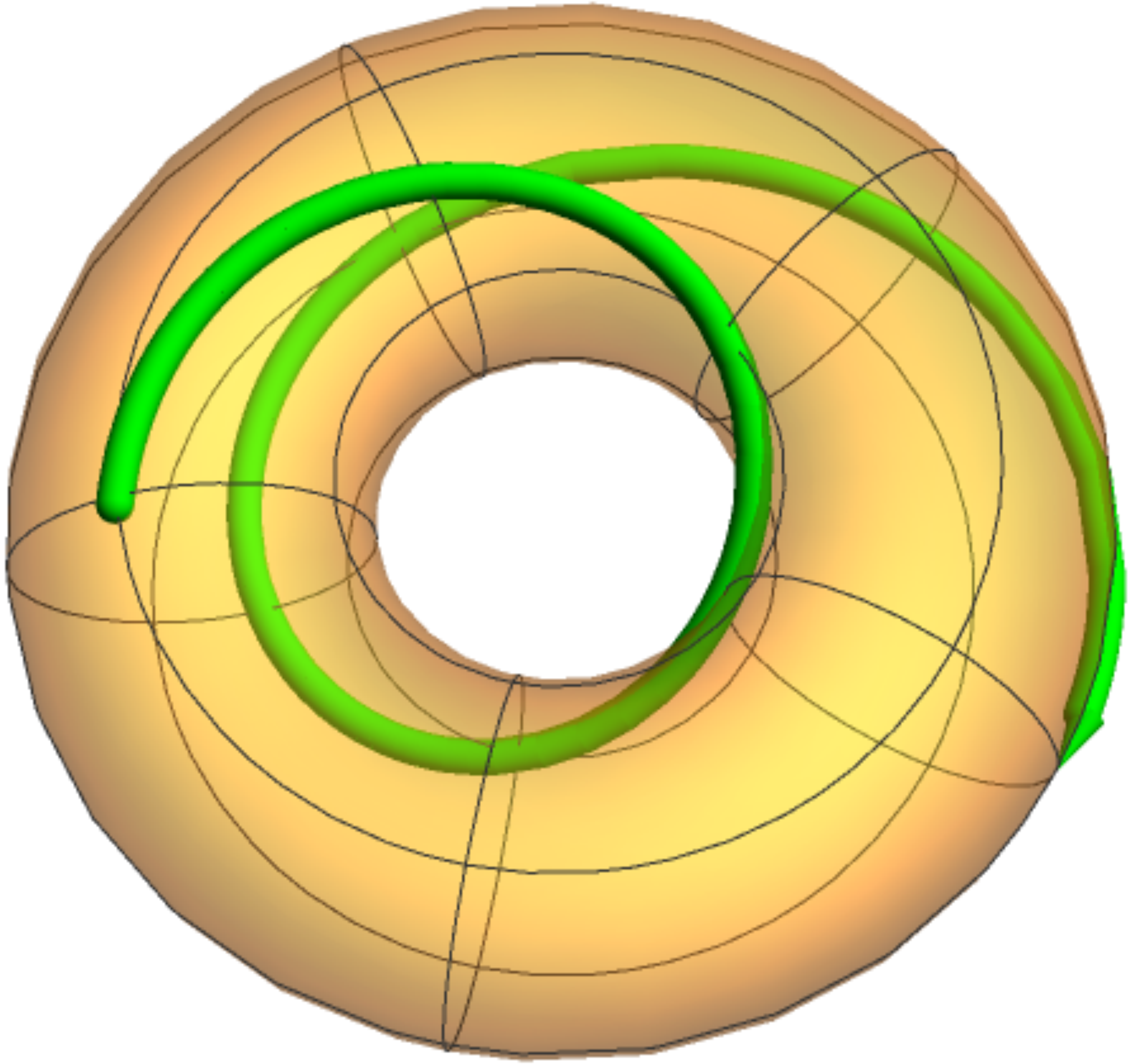

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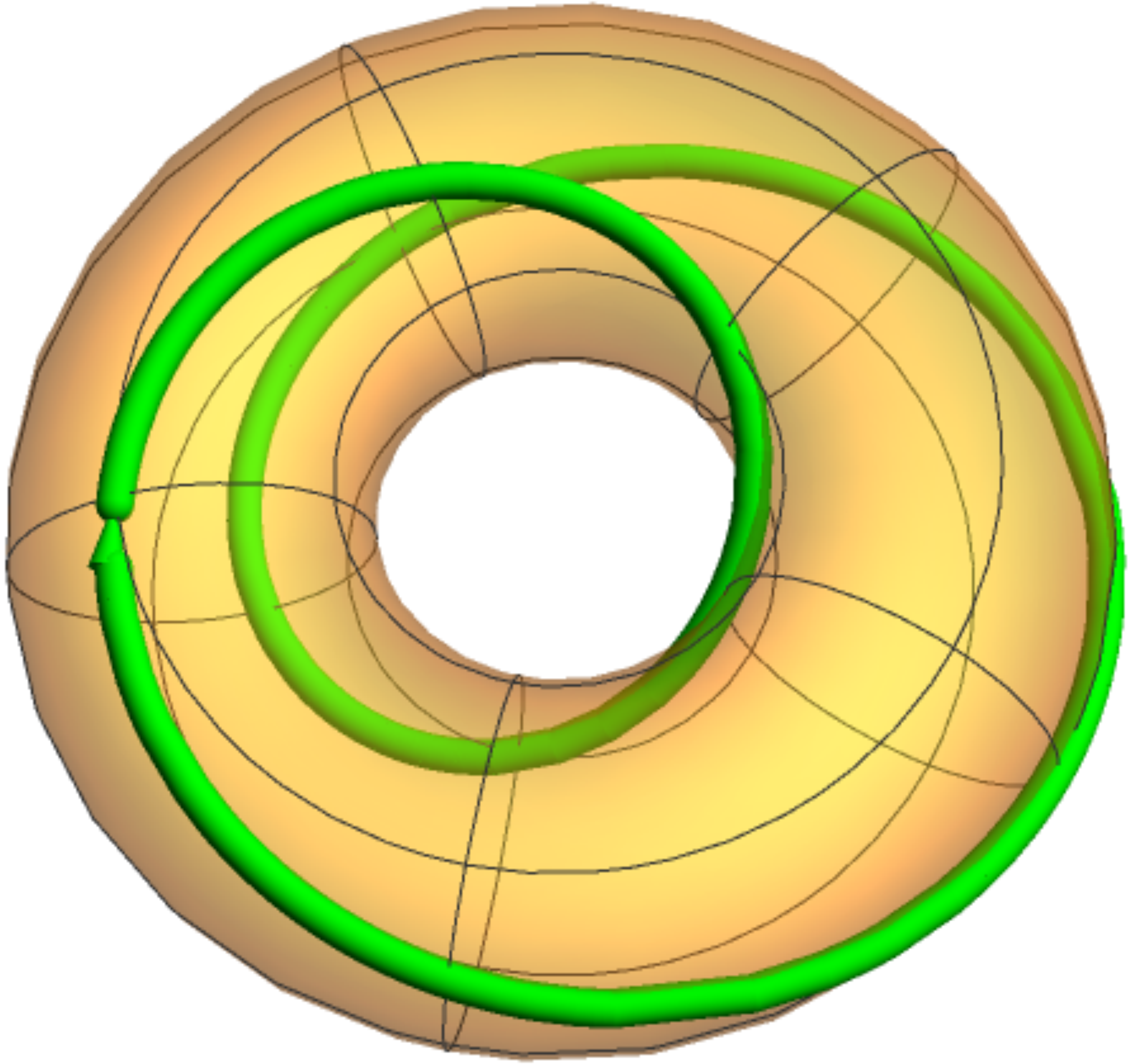

$$\begin{pmatrix}0\\0\end{pmatrix} \quad \begin{pmatrix}1\\2\end{pmatrix} \quad \begin{pmatrix}2\\4\end{pmatrix} \quad \begin{pmatrix}3\\1\end{pmatrix} \quad \begin{pmatrix}4\\3\end{pmatrix}$$

Why care about finite fields and vector spaces over them?

- Course standpoint: They stretch our intuition and provide interesting examples.
- Learning standpoint: They are finite, so are great for concrete calculations.
- Math standpoint: They arise naturally, when you aren't even looking for them.

The “avoiding square products” game

$$6 = 2^1 \times 3^1$$

When multiplying powers of a number, the exponents add:

$$6 \times 2 = (2^1 \times 3^1) \times 2^1 = 2^2 \times 3^1$$

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A natural number is a perfect square when its prime factorization...

has all even exponents.

The "avoiding square products" game

	1	2	3	4	5	6	7	8	9	10
2										
3										
5										
7										

The “avoiding square products” game

	1	2	3	4	5	6	7	8	9	10
2						1				
3						1				
5						0				
7						0				

The “avoiding square products” game

	1	2	3	4	5	6	7	8	9	10
2	0	1	0	2	0	1	0	3	0	1
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multiplying two numbers corresponds to adding these vectors

The “avoiding square products” game

	1	2	3	4	5	6	7	8	9	10
2	0	1	0	0	0	1	0	1	0	1
3	0	0	1	0	0	1	0	0	0	0
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reduce mod 2

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primes form a basis (for numbers up to 10), so any subset of more than 4 vectors is linearly dependent!

Warm-up 2

Which of the following are vector spaces? For those that aren't vector spaces, explain which vector space axiom fails.

a) the vectors in $\mathbb{R}^2$ with non-negative entries

b) the solutions to the equation $x + 2y + 3z = 6$ over the real numbers

c) the solutions to the system of equations

$$\begin{aligned} x + 3y + 4z &= 0 \\ x + 2y + 3z &= 0 \end{aligned}$$

where the variables take values in $\mathbb{R}$

And what if instead they take values in $\mathbb{C}$? Or $\mathbb{Z}/5\mathbb{Z}$?

Warm-up 2

Which of the following are vector spaces? For those that aren't vector spaces, explain which vector space axiom fails.

✗ a) the vectors in $\mathbb{R}^2$ with non-negative entries

not all vectors have additive inverses

✗ b) the solutions to the equation $x + 2y + 3z = 6$ over the real numbers

does not contain a zero vector

✓ c) the solutions to the system of equations

$$\begin{aligned} x + 3y + 4z &= 0 \\ x + 2y + 3z &= 0 \end{aligned}$$

where the variables take values in $\mathbb{R}$

And what if instead they take values in $\mathbb{C}$? Or $\mathbb{Z}/5\mathbb{Z}$?

The solutions to a homogeneous system of linear equations over a field form a vector space!

Because they form a subspace in $\mathbb{F}^n$

Ten Important Terms

A **vector space** is a set V of vectors with the structure of a commutative group under addition, a field $\mathbb{F}$ of scalars, and a multiplication of vectors by scalars that distributes over addition of vectors.

Examples of vector spaces:

$V = \mathbb{F}^n$ and $\mathbb{F}$ the field

$\mathbb{R}^n$ and $\mathbb{C}^n$ $\qquad$ $\mathbb{Q}^n$ $\qquad$ $(\mathbb{Z}/p\mathbb{Z})^n$

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$\mathscr{P}_n$ - real polynomials of degree at most *n*

$$3(x^2 + 5x + 2) \ + 7(x + 1) = 3x^2 + 12x + 13$$

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$\mathscr{P}_n$ - real polynomials of degree at most n

$$3(x^2 + 5x + 2) \ + 7(x + 1) = 3x^2 + 12x + 13$$

$$3\begin{pmatrix}1\\5\\2\end{pmatrix} + 7\begin{pmatrix}0\\1\\1\end{pmatrix} = \begin{pmatrix}3\\12\\13\end{pmatrix}$$

$$\mathscr{P}_n \cong \mathbb{R}^{n+1}$$

isomorphic

What about $\mathscr{P}$ - real polynomials?

It's a vector space, but does not have a finite spanning set.

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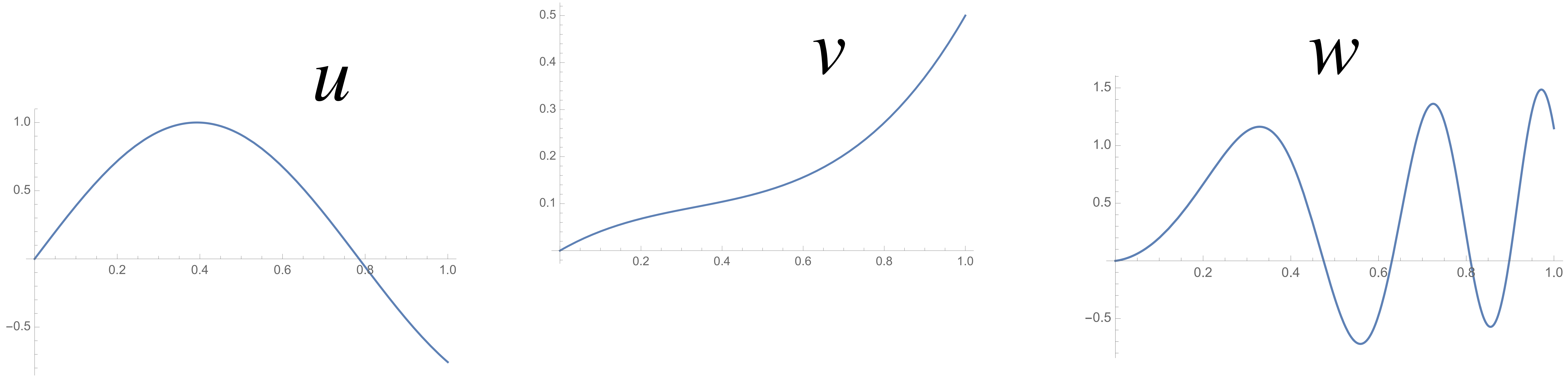
Similarly: continuous functions on [0,1]



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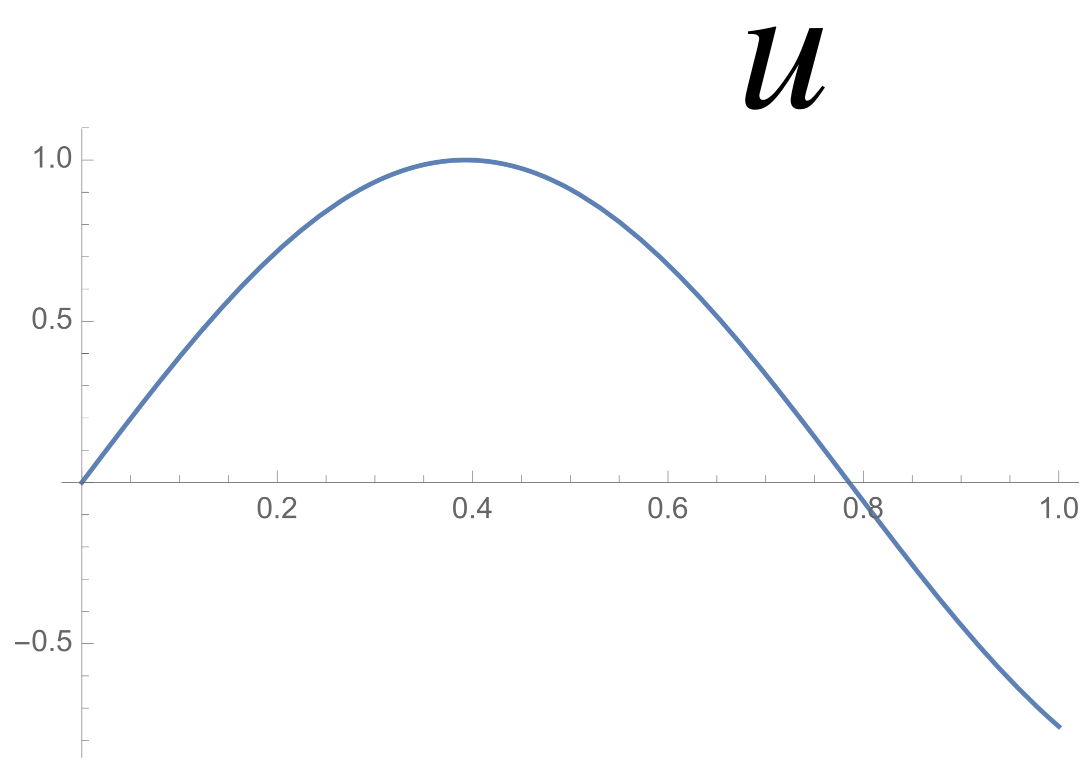

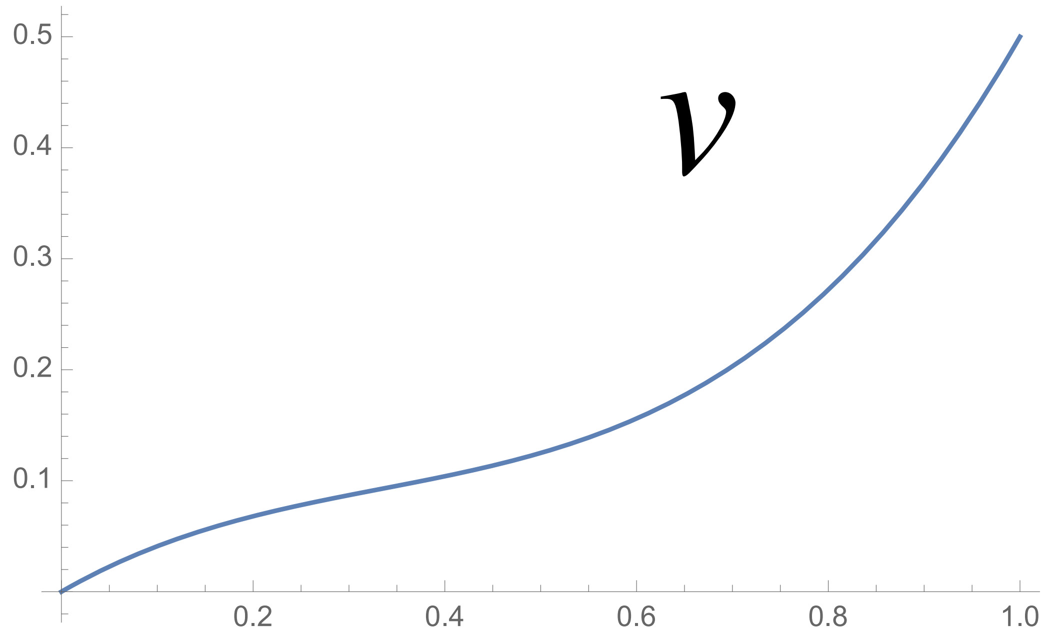

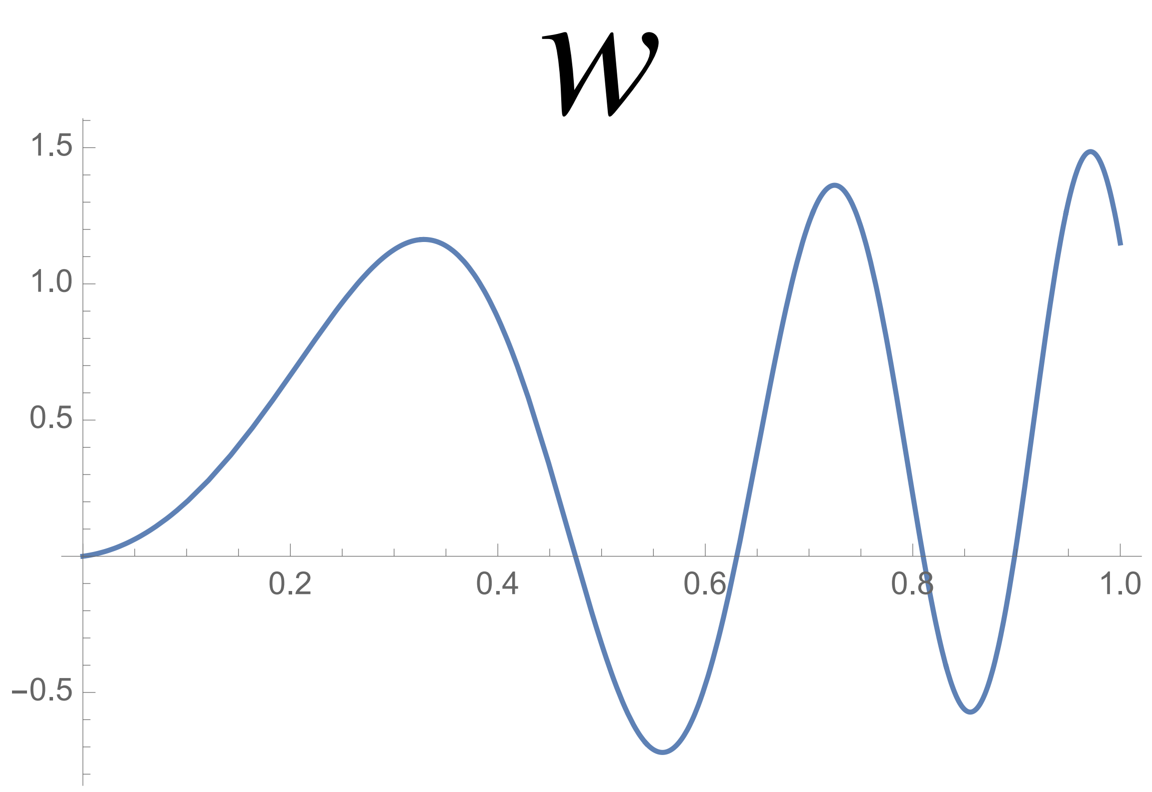

Goal: Classify all vector spaces.

More manageable goal: classify all vector spaces that have a finite spanning set.

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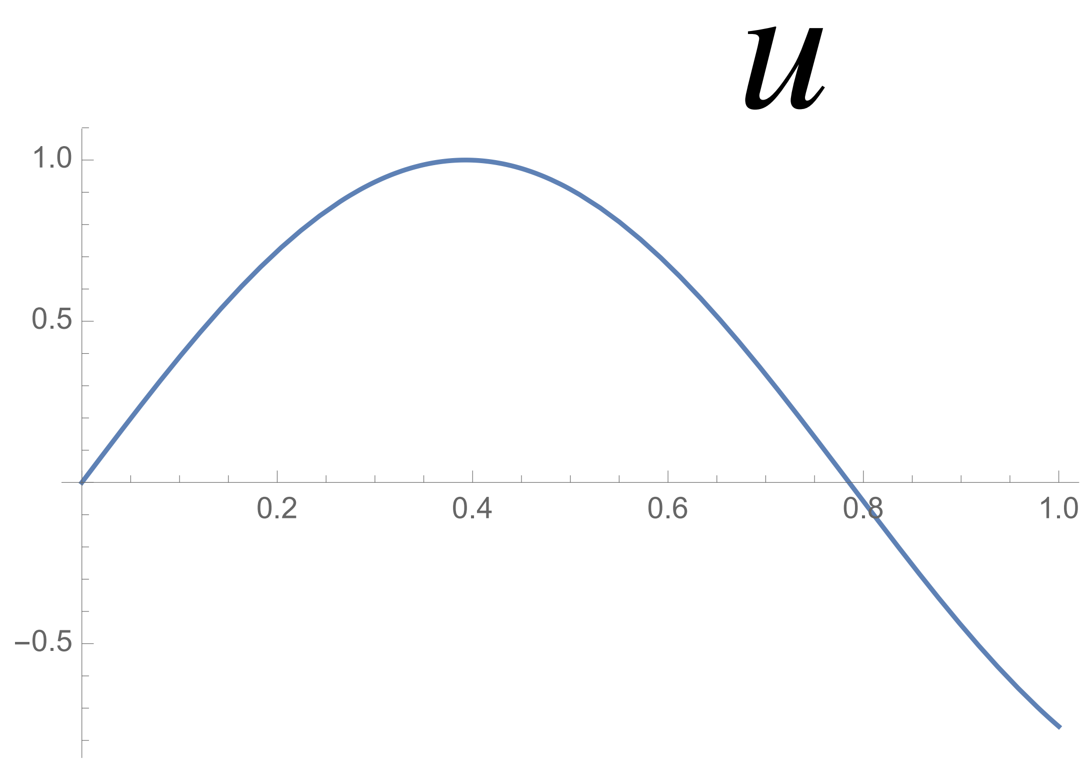
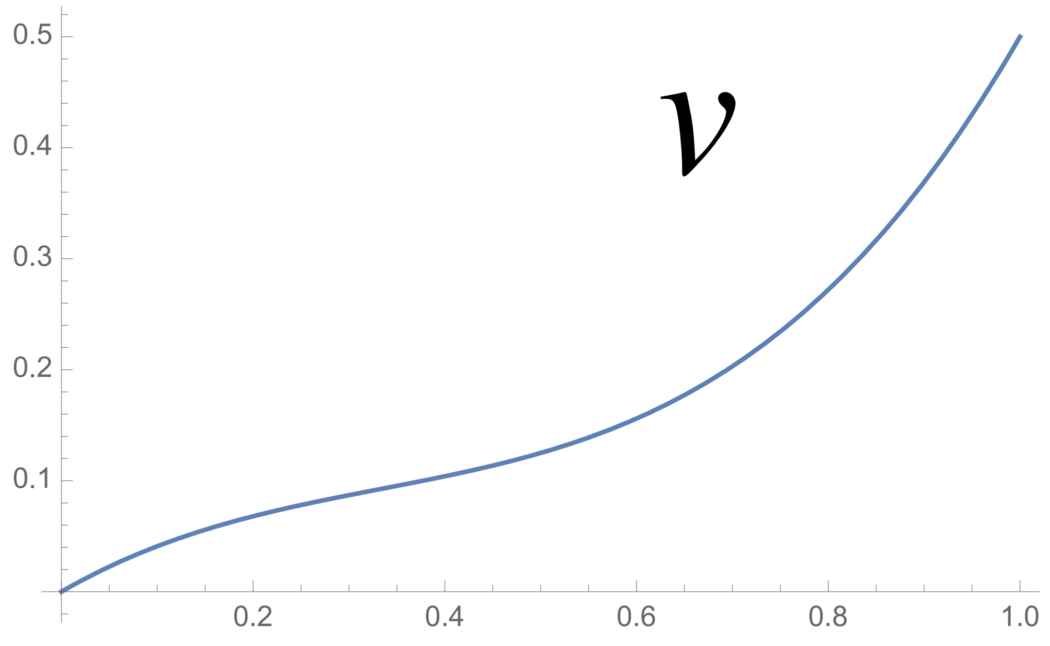
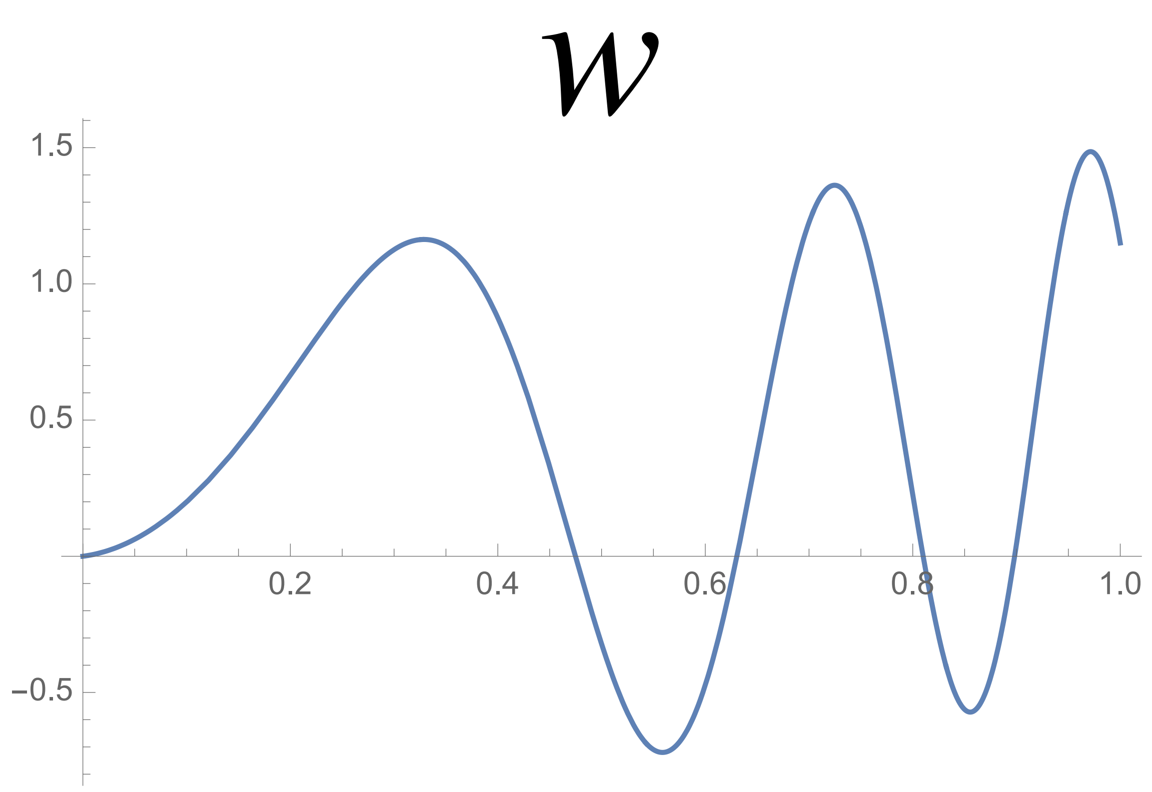






Goal: Classify all vector spaces.

More manageable goal: classify all vector spaces that have a finite spanning set.

Goal: What can we prove about arbitrary vector spaces?

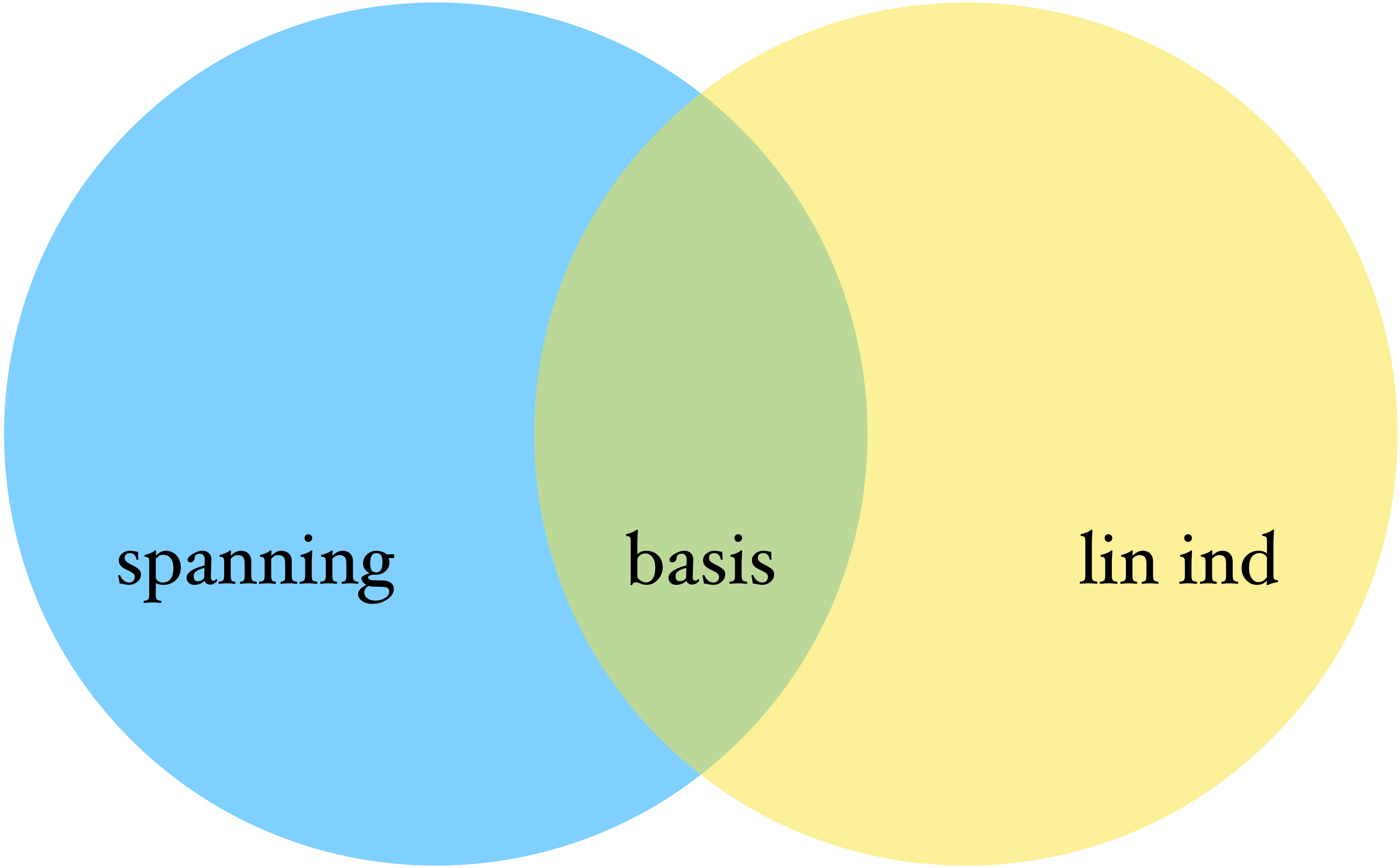
Ten Important Terms

A **vector space** is a set V of vectors with the structure of a commutative group under addition, a field $\mathbb{F}$ of scalars, and a multiplication of vectors by scalars that distributes over addition of vectors.

A **linear combination** of vectors is a finite sum of scalar multiples of the vectors. For vectors $v_1, v_2, \ldots, v_p$, a linear combination of is a sum of form $\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_p v_p = \Sigma_1^p \alpha_k v_k$ where the α_k are scalars. The **span** of a set of vectors in a vector space is the set of all of their linear combinations.

A **linear transformation** is a function between vector spaces over the same field that respects the operations of vector addition and scalar multiplication: $L(\alpha v + \beta w) = \alpha L(v) + \beta L(w)$.

A **basis** of a vector space V is a set of vectors in V such that any vector v in V admits a unique representation as a linear combination of the vectors in the basis. The cardinality of a basis is the **dimension** of a vector space. (*Note: it is a theorem that dimension is well-defined.*)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basis
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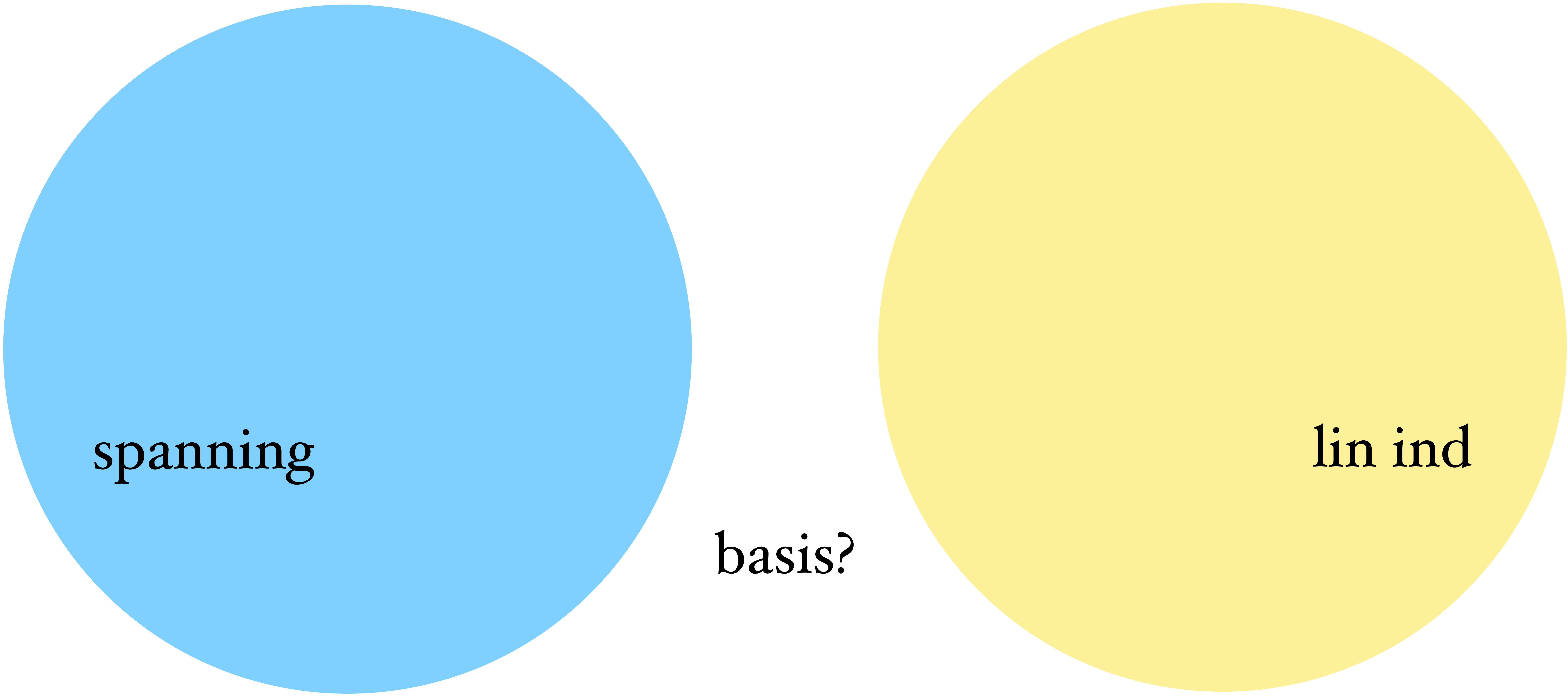

Every vector space has a spanning set (the whole set) and a linearly independents set (the empty set).

Does every vector space have a basis?

vector space definition:

A set of vectors V and a field of scalars $\mathbb{F}$ with operations addition and scaling such that

1. Commutativity: $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$ for all $\mathbf{v}, \mathbf{w} \in V$;
2. Associativity: $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$;
3. Zero vector: there exists a special vector, denoted by $\mathbf{0}$ such that $\mathbf{v} + \mathbf{0} = \mathbf{v}$ for all $\mathbf{v} \in V$;
4. Additive inverse: For every vector $\mathbf{v} \in V$ there exists a vector $\mathbf{w} \in V$ such that $\mathbf{v} + \mathbf{w} = \mathbf{0}$. Such additive inverse is usually denoted as $-\mathbf{v}$;

   The next two properties concern multiplication:

5. Multiplicative identity: $1\mathbf{v} = \mathbf{v}$ for all $\mathbf{v} \in V$;
6. Multiplicative associativity: $(\alpha\beta)\mathbf{v} = \alpha(\beta\mathbf{v})$ for all $\mathbf{v} \in V$ and all scalars α, β;

   And finally, two distributive properties, which connect multiplication and addition:

7. $\alpha(\mathbf{u} + \mathbf{v}) = \alpha\mathbf{u} + \alpha\mathbf{v}$ for all $\mathbf{u}, \mathbf{v} \in V$ and all scalars α;
8. $(\alpha + \beta)\mathbf{v} = \alpha\mathbf{v} + \beta\mathbf{v}$ for all $\mathbf{v} \in V$ and all scalars α, β.

Some results that can be shown for arbitrary vector spaces "from the definition":

The zero vector is unique.

Any vector has a unique inverse.

0v equals the zero vector for any v.

(-1)v equals -v for any v.

Prop. Let V be a vector space. The zero vector is unique.

Want to show: if there are two vector satisfying the zero vector property, they are in fact equal.

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Proof. Suppose that $0, \theta$ each have the zero vector property, so that

$$v + 0 = v \qquad \text{and} \qquad v + \theta = v \qquad \text{for all } v \in V.$$

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Then in particular the following equalities hold:

$$0 + \theta = 0$$

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$$\theta + 0 = 0 + \theta = 0$$

(by commutativity)

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Then in particular the following equalities hold:

$$\theta = \theta + 0 = 0 + \theta = 0$$

So $\theta = 0$, as desired. ∎

Prop. Let V be a vector space. The zero vector is unique.

By contradiction:

Want to show: if there are two vector satisfying the zero vector property, they are in fact equal.

Proof. Suppose that $0, \theta$ each have the zero vector property, so that

$$v + 0 = v \qquad \text{and} \qquad v + \theta = v \qquad \text{for all } v \in V.$$

and $\theta \neq 0$

Then in particular the following equalities hold:

$$\theta = \theta + 0 = 0 + \theta = 0$$

So $\theta = 0$, as desired. ∎

and $\theta \neq 0$, contradiction, so 0 is unique.

Eventually want to show:

Every vector space with a finite spanning set has a (finite) basis.

Any two finite bases for a vector space have the same cardinatlity.
(So dimension is well defined.)

Every finite-dimensional vector space over a field $\mathbb{F}$ is isomorphic to some $\mathbb{F}^n$.

(Upshot: we lose no generality by considering the vector spaces $\mathbb{F}^n$.)